

1) Smart Home Automation System.

A Smart Home automation system controls temp, light and security by using sensor data such as motion detection, room temp and time of day.

Type of Intelligent Agents.

1) Simple Reflex Agent.

* acts only on the current input using predefined rules

Advantages: Fast and Simple

Disadvantage: Cannot learn user preferences

2) Model-Based Agent.

* If maintaining an internal model of the home environment.

* Remembers previous states such as occupied rooms and current device status

Advantage: Better decisions in partially environment

Disadvantage: Requires more memory and processing

3) Goal Based Agent

- * Makes decisions to achieve specific goals
- * Goals include
 - ⇒ Ensure home security
 - ⇒ Maintain home comfortable temp
- * Advantage: Flexible and intelligent planning
- * ~~Dis~~ disadvantage: planning takes more computation.

4) Utility Based Agent

- * Selects the action that provides the highest overall benefit
- * Balances comfort energy savings and security
- * Advantage : produces the best overall performance
- * Disadvantage : utility calculation is more complex

Robot in an Unknown Maze

A robot is placed in an unknown maze and must find the exit without knowing the layout beforehand. It can use uniform Cost search (UCS) and Iterative Deepening search (IDS) to explore the maze

Uniform Cost Search

- ✧ Expands the node with the lowest Cumulative Path cost
- ✧ Guarantees the least Cost Path

Advantage

- ✧ complete
- ✧ Finds the optimal solution when costs differ

Disadvantage

- ✧ Uses large memory
- ✧ Slower for large search spaces

Complexity

- ✧ Time : exponential
- ✧ Space : Exponential

Iterative Deepening Search (IDS)

- * Repeatedly performs Depth-first Search with Increasing depth limits
- * Combines DFS's low memory with BFS's completeness

Advantages

- * Complete
- * Very low memory usage
- * Suitable for large search spaces.

Disadvantages

- * Repeats node expansions at each depth
- * Not optimal when path costs vary

Complexity

- * Time $O(b^d)$
- * Space $O(bd)$.